Chapter 4, Section 1

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1. Let X be a curve, and let $P \in X$ be a point. Then there exists a nonconstant rational function $f \in K(X)$, which is regular everywhere except at P.

Proof. For large $n > 0$, we have $\mathrm{H}^0(\mathcal{O}_X(nP)) \neq 0$ (1.3.2). This implies the existence of an effective divisor D' such that $D' - nP = (f)$, where f is a rational function and (f) is the divisor of zeros of f. Since D' is effective, the only pole of f is at P of order at most n. □

5. For an effective divisor D on a curve X of genus g, show that $\dim |D| \leq \deg D$. Furthermore, equality holds if and only if $D = 0$ or $g = 0$.

Proof. The Riemann-Roch theorem gives

$$\ell(D) - 1 = \ell(K - D) + \deg D - g,$$

so it suffices to show $\ell(K - D) \leq g$ and equality holds if and only if $D = 0$ or $g = 0$.

If D is effective, then we have an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

Tensoring by $\mathcal{O}_X(K)$ and taking global sections gives an injection of vector spaces $\mathrm{H}^0(\mathcal{O}_X(K - D)) \to \mathrm{H}^0(\mathcal{O}_X(K))$, so we conclude that $\ell(K - D) \leq \ell(K) = g$.

Now, we have $\dim |D| = \deg D$ if and only if $\ell(K - D) = \ell(K)$, and it is clear this equality holds if either $D = 0$ or $g = 0$. Conversely, suppose $D \neq 0$ and $g \neq 0$. We will show $\ell(K - D) < \ell(K)$. Since D is effective, $d = \deg D > 0$, so we proceed by induction on d. If $D = P$ for a point $P \in X$, then $\ell(P) = 1$ because X is not rational. Thus, Riemann-Roch gives $\ell(K - P) = g - 1$.

Assume the statement for $d > 1$, and let D be an effective divisor of degree $d + 1$. Let P be a point in the support of D so that $D' = D - P$ is an effective divisor of degree d. By the inductive hypothesis, we have $\ell(K - D') < \ell(K)$. Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_P \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(K - D')$ and taking global sections gives an inclusion of vector spaces $\mathrm{H}^0(\mathcal{O}_X(K - D)) \to \mathrm{H}^0(\mathcal{O}_X(K - D'))$. Thus, $\ell(K - D) \leq \ell(K - D')$, so combining inequalities, we get

$$\ell(K - D) \leq \ell(K - D') < \ell(K).$$

□

6. Let X be a curve of genus g. Show that there is a finite morphism $f : X \to \mathbb{P}^1$ of degree $\leq g + 1$.

Proof. Fix a point $P \in X$, and consider the divisor $D = (g + 1)P$. By Riemann-Roch, $\ell(D) > 1$, which implies the existence of a nonconstant rational function $f \in K(X)$ with poles no worse than D. This means we can write $f = g/h$ such that the order of vanishing of h at P is at most $g + 1$, and h has no other zeros. The inclusion of fields $k(f) \subseteq K(X)$ induces a finite morphism $f : X \to \mathbb{P}^1$. Now, the degree of $f^{-1}(0)$ is at most $g + 1$ points, so f has degree at most $g + 1$ by (II, 6.9). □

7. A curve X is called *hyperelliptic* if $g \geq 2$ and there exists a finite morphism $f : X \to \mathbb{P}^1$ of degree 2.

(a) If X is a curve of genus $g = 2$, show that the canonical divisor defines a complete linear system $|K|$ of degree 2 and dimension 1, without basepoints. Use (II, 7.8.1) to conclude that X is hyperelliptic.

(b) Show that the curves constructed in (1.1.1) all admit a morphism of degree 2 to $\mathbb{P}^1$. Thus, there exist hyperelliptic curves of any genus $g \geq 2$.

Proof.

(a) By Riemann-Roch, $\deg K = 2g - 2 = 2$ and $\dim |K| = \ell(K) - 1 = 1$. For any $P \in X$, consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(K - P) \longrightarrow \mathcal{O}_X(K) \longrightarrow k(P) \longrightarrow 0.$$

The linear system $|K|$ is basepoint free if and only if $\mathrm{H}^0(\mathcal{O}_X(K)) \to \mathrm{H}^0(\mathcal{O}_P)$ is surjective for every $P \in X$. Taking the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(K - P)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(K)) \longrightarrow \mathrm{H}^0(P, \mathcal{O}_P).$$

Since $\ell(K) = 2$ and $h^0(P, \mathcal{O}_P) = 1$, it is enough to show $h^0(X, \mathcal{O}_X(K - P)) = 1$. Again, by the Riemann-Roch theorem, $\ell(K - P) = \ell(P)$, and $\ell(P) = 1$ because X is not rational. Hence, $\ell(K - P) = 1$.

(b) If D is a divisor of type (a, b) with $a, b > 0$, then D is very ample, so by Bertini's theorem there exists an irreducible nonsingular curve $C \subseteq Q = \mathbb{P}^1 \times \mathbb{P}^1$ that is linearly equivalent to D and has type (a, b) as a divisor in Q. Also, $a = \deg p_1\big|_C$ and $b = \deg p_2\big|_C$. Suppose $(a, b) = (g + 1, 2)$. By (III, Ex. 5.6), C has genus $= (g + 1 - 1)(2 - 1) = g$, and $p_2\big|_C : C \to \mathbb{P}^1$ is a morphism of degree 2. Hence, X is a hyperelliptic curve of genus g.

□